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Surrogates for Monte Carlo uncertainty propagation with twenty model runs

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***Abstract*— Monte Carlo propagation of distributions needs far more evaluations of the measurement model than an expensive simulation allows, so the model is replaced by a surrogate trained on about twenty runs. We compare surrogate strategies on four test models, including a Gaussian plume model of an ammonia release, over 30 paired repetitions, and score the propagated mean, 95th percentile and exceedance probability against a reference Monte Carlo evaluation. The kernel specification of a Gaussian process changed the error up to eighteen-fold; one missing hyperparameter produced an apparent failure that an earlier analysis had attributed to domain size. Logarithmic transformation of positive variables reduced the error up to eightfold. Committees of different surrogates beat a well-specified Gaussian process only where one member matched the response structure, and only with cross-validation error weighting. Adaptive selection of runs did not help consistently. We derive practical rules from these results.**

***Keywords*—Measurement uncertainty; Monte Carlo method; Surrogate model; Gaussian process; Computer experiments**

I. INTRODUCTION

Evaluating measurement uncertainty by the propagation of distributions, following Supplement 1 to the Guide to the Expression of Uncertainty in Measurement (GUM S1) [1], [2], requires the measurement model $Y = f(\mathbf{X})$ to be evaluated once per Monte Carlo trial, typically 10^5 to 10^6 times [3], [4]. For many quantities of practical interest, f is itself a numerical simulation. One example is the concentration downstream of an accidental ammonia release [5]: a single computational fluid dynamics run takes hours to days, and a practical budget is often fewer than twenty runs.

The usual remedy is a *surrogate* (emulator) trained on those runs and used in place of f inside the Monte Carlo loop [6]–[8]. The surrogate's error then becomes a contribution to the uncertainty budget [9]. Gaussian process (GP) regression [10], [11] is the standard surrogate; a common guideline asks for about ten runs per input [12], which many applications cannot afford, and results are known to depend on modelling choices [13]. Several alternatives have been proposed for small budgets: weighted committees of different surrogates [14], [15], adaptive selection of runs [16] and, recently, spectral measures of committee disagreement [17].

An earlier analysis of such a scheme (heterogeneous committee, spectral acquisition, dynamic model management) reported that a passively sampled GP fails catastrophically on large input domains and that the committee recovers accuracy. This contribution tests which of those conclusions survive a controlled comparison with correctly specified baselines, on models chosen to include cases where no committee member matches the response, and scored on the quantities a GUM S1 evaluation reports.

Table I. Test models; d: number of inputs.

Model	d	Inputs; output
Physics	3	$[-200, 200]^3$
Plume	5	release, wind, height, distance, offset; concentration in mg/m^3
Borehole	8	standard ranges; flow in m^3/yr
Ishigami	3	$[-\pi, \pi]^3$

II. PROPOSED CONTRIBUTION

A. Test models

Four models with uniform input distributions are used (Table I). *Physics* is the benchmark of the earlier analysis; its dominant quadratic term lies exactly in the class of one committee member. *Plume* is the ground-level concentration of a continuous ammonia release from a Gaussian plume model with Briggs rural dispersion coefficients for neutral stability and ground reflection. Its output spans three orders of magnitude and has a 12 % probability of exceeding the level immediately dangerous to life or health (IDLH) for ammonia, 210 mg/m^3 (about 300 ppm). *Borehole* (water flow) and *Ishigami* are standard test functions for computer experiments. No committee member matches the last three.

B. Strategies

Every strategy uses $N = 20$ runs of the model. The single-GP strategies use a 20-point Latin hypercube with five kernel specifications: the original kernel of the earlier analysis (squared exponential with one length scale and *no* amplitude hyperparameter), with an amplitude added, with one length scale per input (automatic relevance determination, ARD), with both, and a Matérn-5/2 ARD kernel with amplitude. A

further variant fits the latter to the logarithm of the output, with logarithmic transformation of inputs whose range is strictly positive (plume and borehole only). The adaptive GP starts from five Latin hypercube runs and adds 15 at its largest posterior standard deviation [16].

The committee has nine members: two GPs (with amplitude and ARD), polynomial chaos expansions (PCE) [18], [19] of degrees 2 and 3, two random forests, a support vector regressor and two neural networks. Its members are combined either uniformly or with weights

$$w_k \propto \left[\mathrm{MSE}_{\mathrm{CV},k}\,(1+\pi_k)\right]^{-1}, \qquad (1)$$

where $\mathrm{MSE}_{\mathrm{CV},k}$ is the three-fold cross-validation error of member k and $\pi_k \geq 0$ penalises predictions below a known physical bound [14], [15]. The committee is trained either on the 20-point design (passive) or on five runs plus 15 chosen where its weighted members disagree most, scored by their variance or by the spectral measure UEigV [17] (adaptive).

C. Evaluation

Each strategy is run with 30 seeds. Seeds are shared, so passive strategies use the identical design and adaptive strategies the identical start. The trained surrogate propagates $3 \cdot 10^4$ input draws, and the propagated mean, standard deviation (SD), 95th percentile $y_{0.95}$ and exceedance probability $P(Y > y_{0.95})$ are compared with a reference Monte Carlo evaluation of the true model ($M = 2 \cdot 10^5$). Accuracy over the input domain is the root mean square error (RMSE) on 5000 test points divided by the output SD (normalised RMSE; 1 corresponds to a constant predictor). Strategies are compared by a two-sided Wilcoxon signed-rank test on paired log normalised RMSE. Code and every per-seed record are openly available [20].

III. PRELIMINARY RESULTS AND POTENTIAL APPLICATIONS

A. Kernel specification and variable transformation

Table II shows that the specification of a single GP matters more than any other choice examined. On the physics model the original kernel reaches 0.59, close to a constant predictor; adding the amplitude hyperparameter alone reduces this to 0.033 (all 30 seeds improve, $p < 10^{-8}$). The catastrophic GP failure reported earlier is therefore a specification artefact (Fig. 1): with the original kernel the normalised RMSE is 0.60 at every domain half-width from 10 to 200, whereas a Matérn-5/2 ARD GP with amplitude reaches 0.17 at 10 and 0.08 from 50 to 200. The committee remains far more accurate on this model, but because of its degree-2 PCE member, not because the GP fails. On the borehole model ARD matters (0.30 to 0.09). Logarithmic transformation then gives the largest single improvement: from 0.089 to 0.011 on the borehole model and from 0.60 to 0.42 on the plume (30 and 27 of 30 seeds improve). No GP does better than 0.9 on the Ishigami function, for which the ten-runs-per-input guideline [12] would ask for 30 runs.

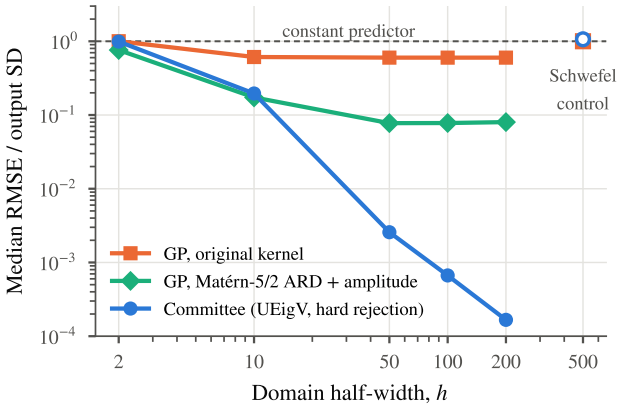


Fig. 1. Physics model on $[-h, h]^3$: median normalised RMSE over 30 seeds against the domain half-width. Open markers: Schwefel function on $[-500, 500]^3$. Dashed line: constant predictor.

Table II. Median normalised RMSE of a single GP on a 20-point Latin hypercube, 30 seeds. Amp.: amplitude hyperparameter; Matérn: Matérn-5/2 kernel.

Kernel	Phys.	Plume	Boreh.	Ishig.
Original (no amp.)	0.591	0.602	0.295	0.956
+ amp.	0.033	0.590	0.247	0.970
+ ARD	0.608	0.632	0.129	1.01
+ amp. + ARD	0.036	0.596	0.085	0.896
Matérn, amp. + ARD	0.079	0.597	0.089	0.902
on log variables	–	0.418	0.011	–

B. Committees and adaptive runs

Fig. 2 compares all strategies. A committee outperforms the best single GP only on the physics model, by more than two orders of magnitude, and only because the degree-2 PCE represents its dominant term exactly; the weights then concentrate on that member. On the borehole model the CV-weighted committee is 2.3 times worse than the Matérn-5/2 GP on the same design (0 of 30 seeds better), and on the plume and Ishigami models the two tie. Aggregation matters wherever a committee helps: uniform averaging is worse than CV weighting on the physics (by a factor of about 2000), plume and borehole models ($p \leq 0.023$) and never better. Adaptive selection of runs gave no consistent benefit. For the single GP it was never better and was worse on the plume ($p = 0.008$). For the committee it helped on the physics and borehole models (median ratios 0.80 and 0.61), hurt on the plume (1.25) and made no difference on the Ishigami function. Spectral (UEigV) acquisition was worse than variance acquisition on three models (on the borehole model by a factor of 1.7, $p < 10^{-7}$) and better only on the plume (0.90, $p = 0.009$).

C. Propagated quantities

Table III gives the quantities a GUM S1 evaluation reports, and they follow the normalised RMSE. With the best strategy for each model, the 95th percentile is within 0.41 % (physics), 6.9 % (plume), 0.60 % (borehole) and 16 % (Ishigami) of the reference, and $P(Y > y_{0.95})$ within

Table III. Propagation error against the reference Monte Carlo evaluation, median over 30 seeds: relative error (%) of the mean and of $y_{0.95}$, and absolute error (×100) of $P(Y > y_{0.95}) = 0.05$.

	Physics			Plume			Borehole			Ishigami		
Strategy	mean	$y_{0.95}$	P	mean	$y_{0.95}$	P	mean	$y_{0.95}$	P	mean	$y_{0.95}$	P
GP, original kernel	8.90	21.1	4.86	10.4	20.4	3.29	1.60	13.0	4.25	10.3	34.6	4.73
GP, Matérn-5/2 ARD	1.36	3.88	1.37	9.68	17.7	2.87	0.66	2.45	0.91	**8.48**	**16.0**	3.61
GP, log variables	–	–	–	**4.74**	**6.94**	**0.76**	**0.17**	**0.60**	**0.18**	–	–	–
Committee, CV-weighted	**0.21**	**0.41**	**0.15**	8.75	23.1	3.15	1.06	10.9	3.05	11.0	28.7	**2.89**
Committee, adaptive	**0.21**	**0.41**	**0.15**	55.1	20.5	5.39	3.30	1.41	0.49	15.9	34.9	4.40

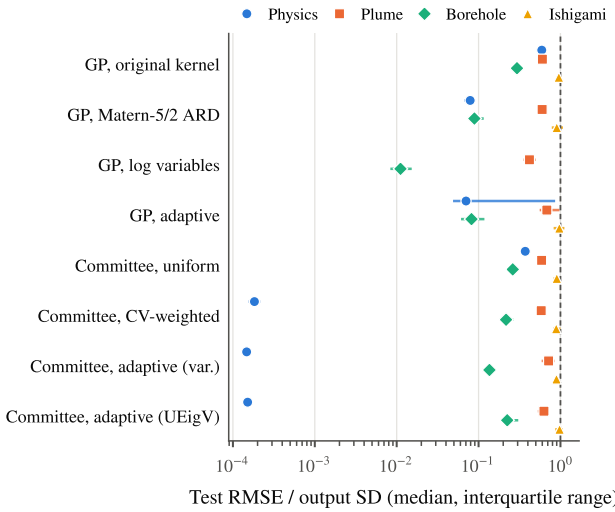


Fig. 2. Normalised RMSE of each strategy on each test model: median (marker) and interquartile range (bar) over 30 seeds. Dashed line: constant predictor. Committees use GP members with amplitude and ARD.

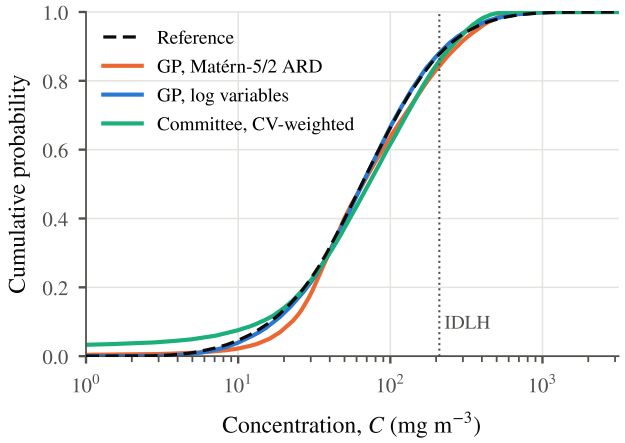


Fig. 3. Plume model, first seed: distribution of the ground-level concentration propagated through three surrogates fitted to the same 20 runs, and the reference Monte Carlo distribution. Dotted: IDLH level for ammonia.

0.0015, 0.0076, 0.0018 and 0.029 of 0.05. Fig. 3 shows the plume case for the first seed: with log variables the GP reproduces the reference distribution over three orders of magnitude and gives $P(C > 210\,\mathrm{mg/m^3}) = 0.119$ against a reference 0.123, compared with 0.158 for the GP on raw variables and 0.142 for the committee. Adaptive runs can mislead: the adaptive committee estimated the plume mean with a median error of 55 %. The member envelope of a committee, read as a model-form interval for $P(Y > y_{0.95})$, contained the reference value in between 0 % and 100 % of seeds depending on model and strategy; it cannot replace a calibrated interval [21], [22].

D. Potential applications

The results suggest rules for surrogate-based GUM S1 evaluations with a budget of about twenty runs, such as a dispersion study in a risk assessment. *Specify the GP fully*: an amplitude hyperparameter and one length scale per input; check the cross-validation error against the output SD before propagating. *Transform the variables* when the model is multiplicative or positive, for example concentrations and flows. *Use a committee only with error weighting*, and expect a gain only when a member can represent the dominant structure; otherwise a well-specified GP is as good or better. *Do not assume that adaptive selection of runs helps* at this budget; a space-filling design is a safe default. *Report the surrogate's error* as a component of the uncertainty budget, estimated from held-out or cross-validation error rather than from the spread of committee members.

IV. CONCLUSIONS

For Monte Carlo uncertainty propagation with twenty runs of an expensive model, the kernel specification of a GP surrogate and the transformation of the variables mattered more than the choice between single surrogates, committees and adaptive designs. A failure of GP surrogates reported in an earlier analysis was caused by one missing hyperparameter. Committees helped only when one member matched the response structure and only with cross-validation error weighting, and adaptive selection of runs was not reliably beneficial. The final paper will add a CFD-based dispersion case and calibrated (conformal) intervals for the surrogate contribution.

AUTHORS' CONTRIBUTION

Yugam Jinesh Vora: conceptualisation, methodology, software, investigation, formal analysis, visualisation, writing – original draft. Wee Hoe Ng: conceptualisation, supervision, writing – review and editing.

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